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(54) **THICK STEEL PLATE ASSEMBLY WELDED BY LASER WELDING AND LASER WELDING METHOD THEREOF**

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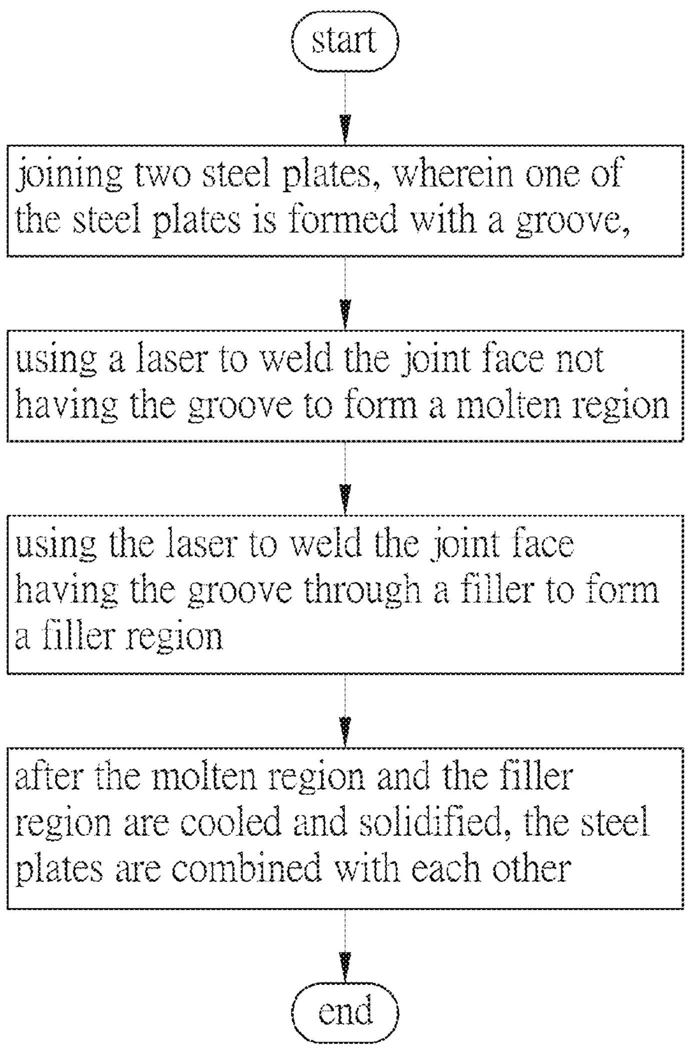
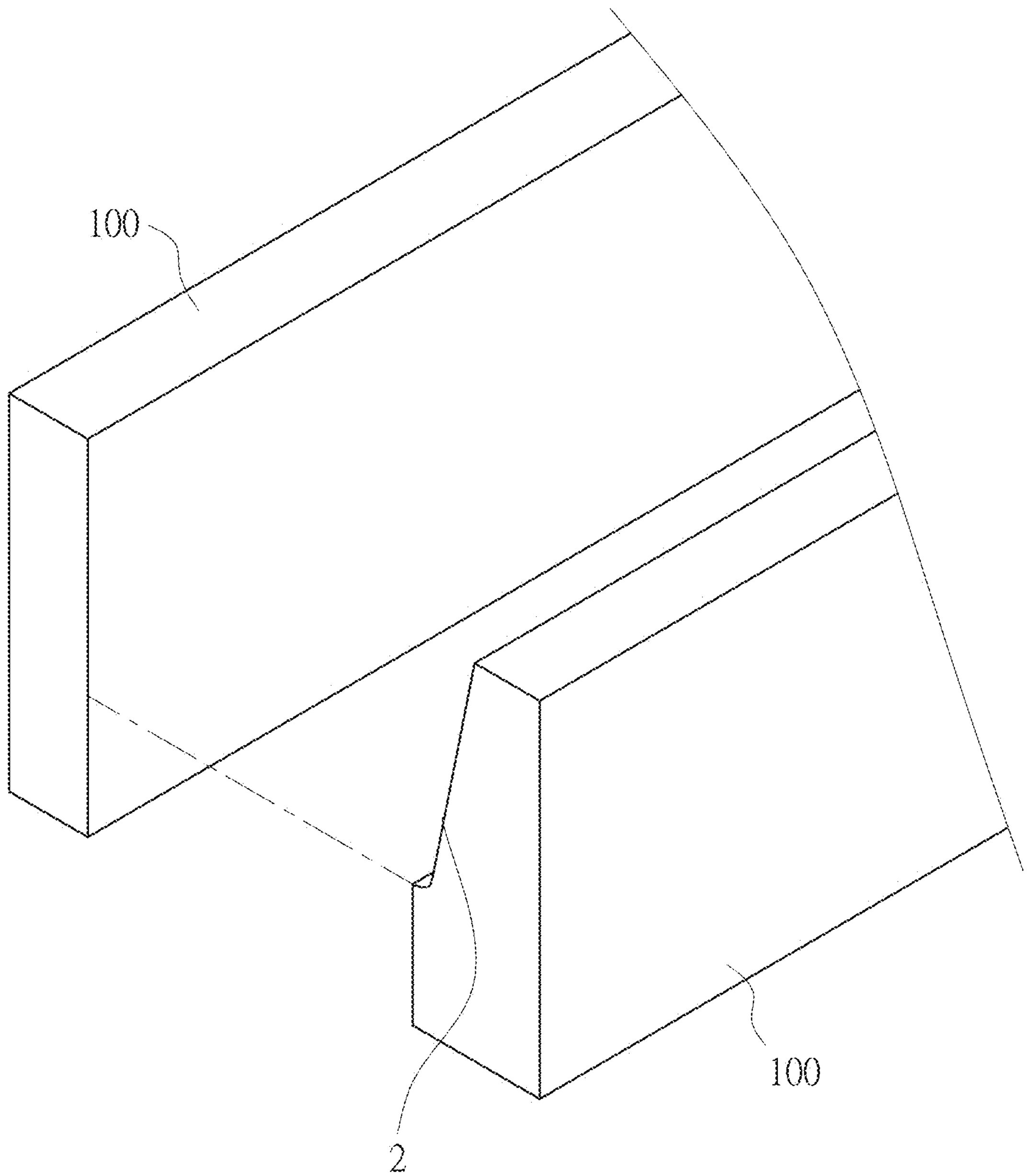
(57) **ABSTRACT**

A thick steel plate assembly welded by laser welding and a laser welding method thereof are disclosed. The thick steel plate assembly comprises two steel plates and a filler region. A joint face of the steel plates extends along a joint line. At least one portion of the joint face of one of the thick plate steels is formed with a groove along a grooving line. The filler region is formed by feeding a filler into the groove and using a laser to weld the joint face having the groove through the filler. After the filler region is cooled and solidified, the thick steel plates are combined with each other. A thickness of each filler layer in the filler region is between 4 and 6 mm. Thus, the process can be simplified greatly and the cost can be reduced.





F I G . 1

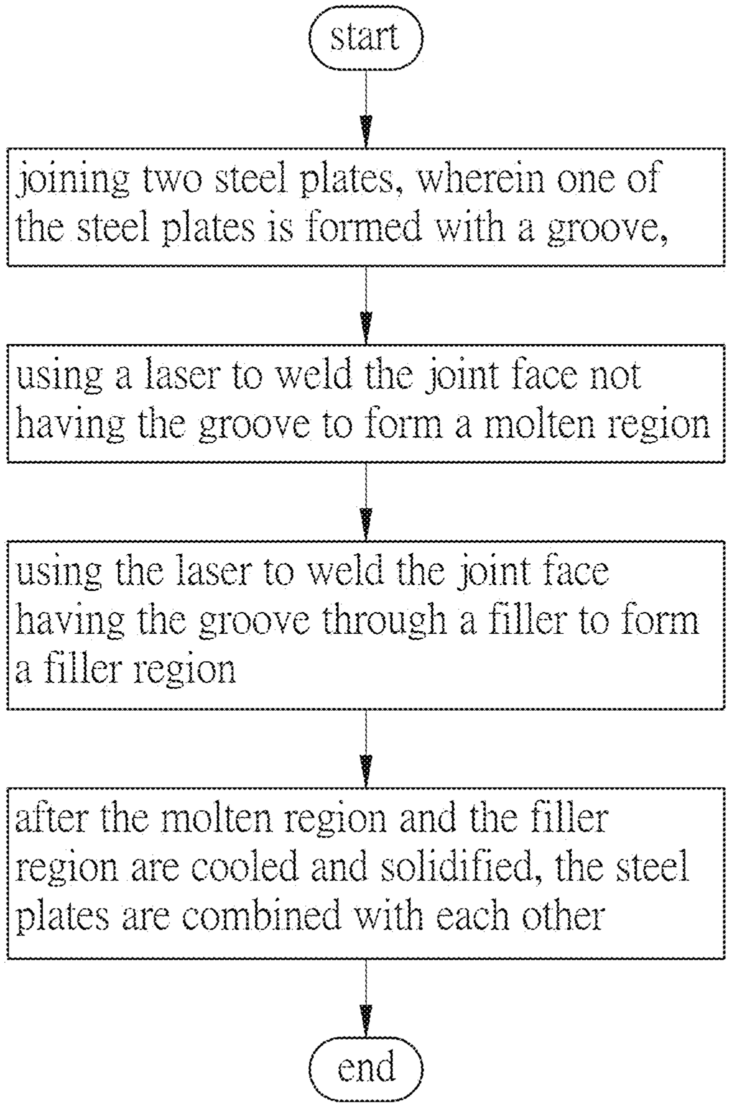
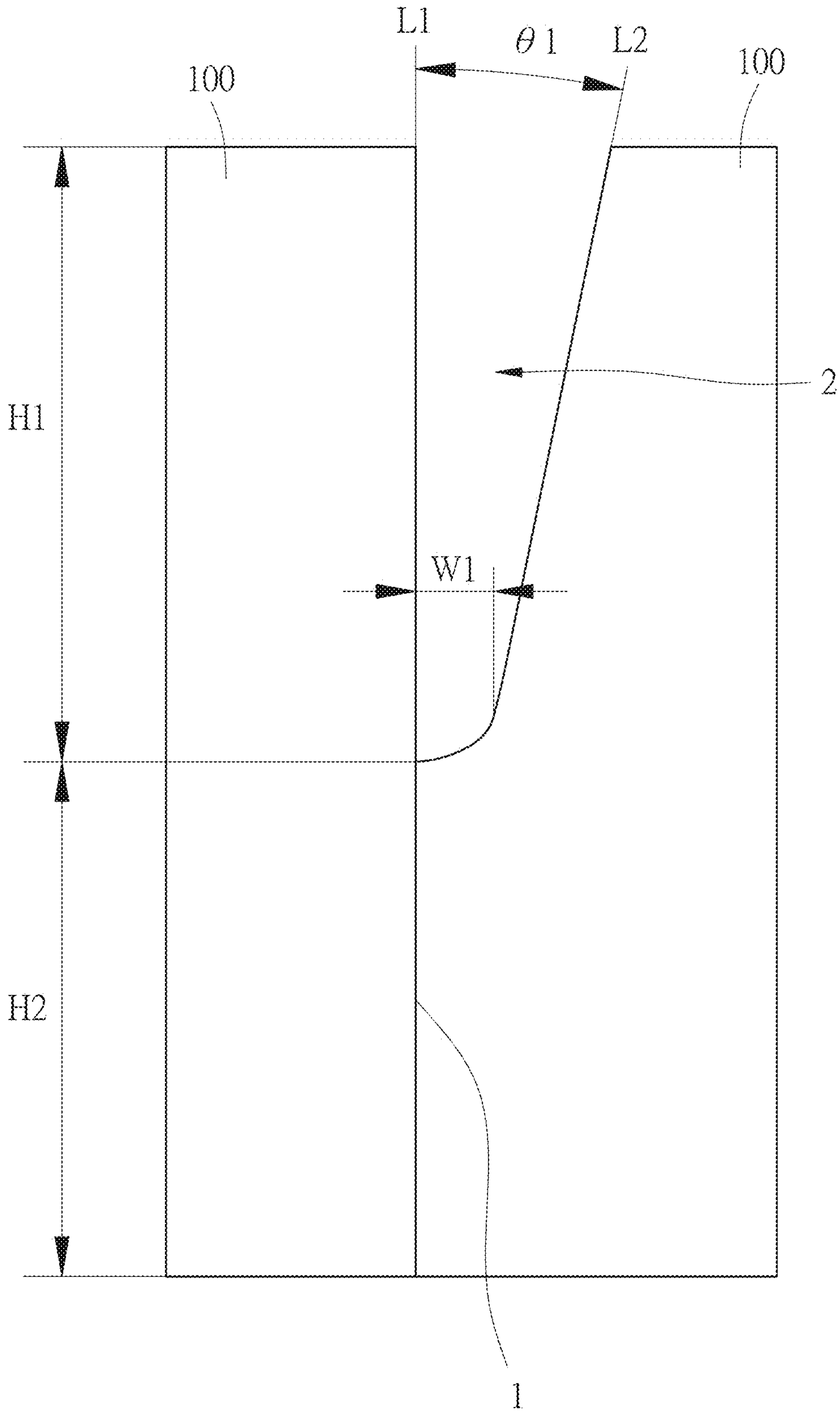
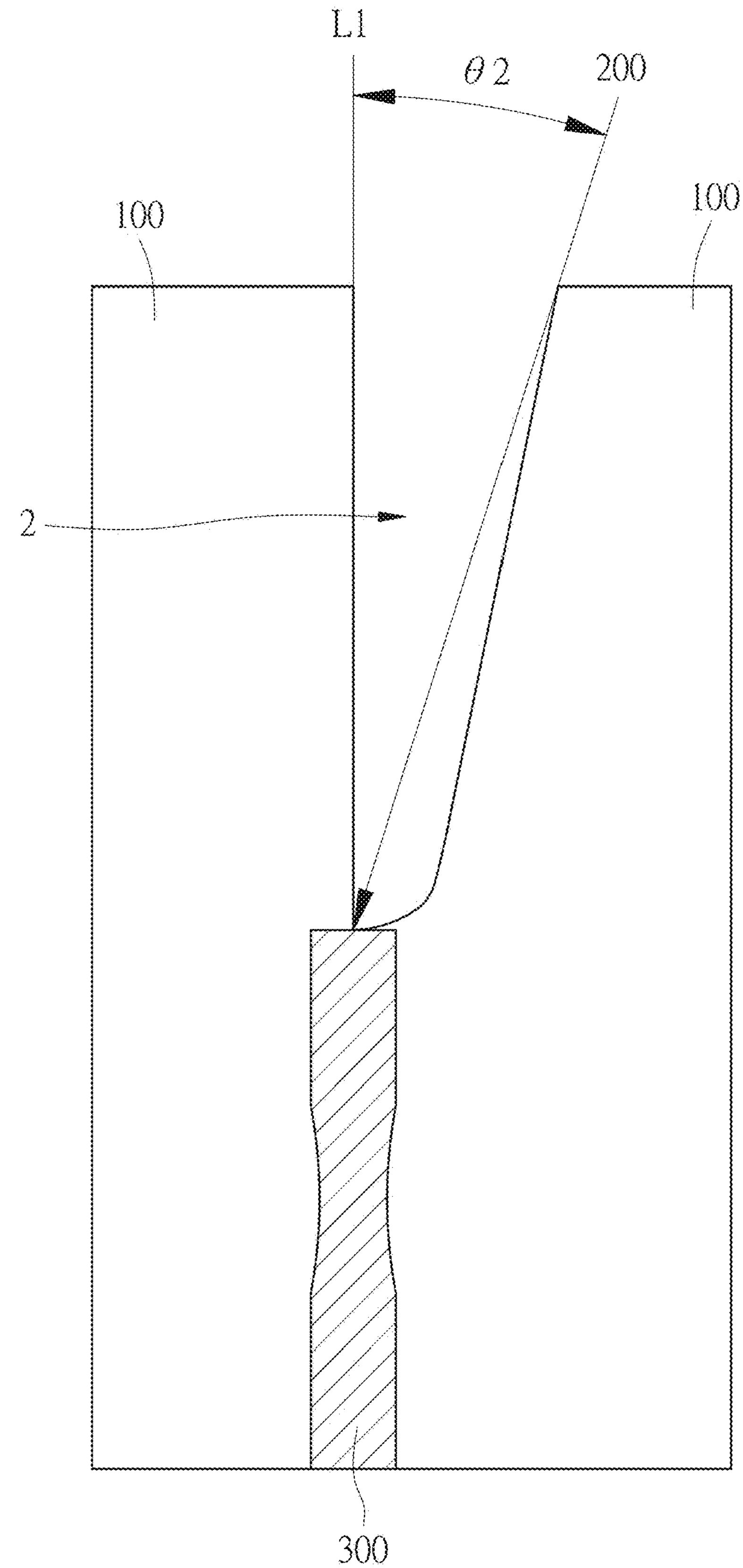

start
joining two steel plates, wherein one of the steel plates is formed with a groove,
using a laser to weld the joint face not having the groove to form a molten region
using the laser to weld the joint face having the groove through a filler to form a filler region
after the molten region and the filler region are cooled and solidified, the steel plates are combined with each other
end

FIG. 2



F I G . 3



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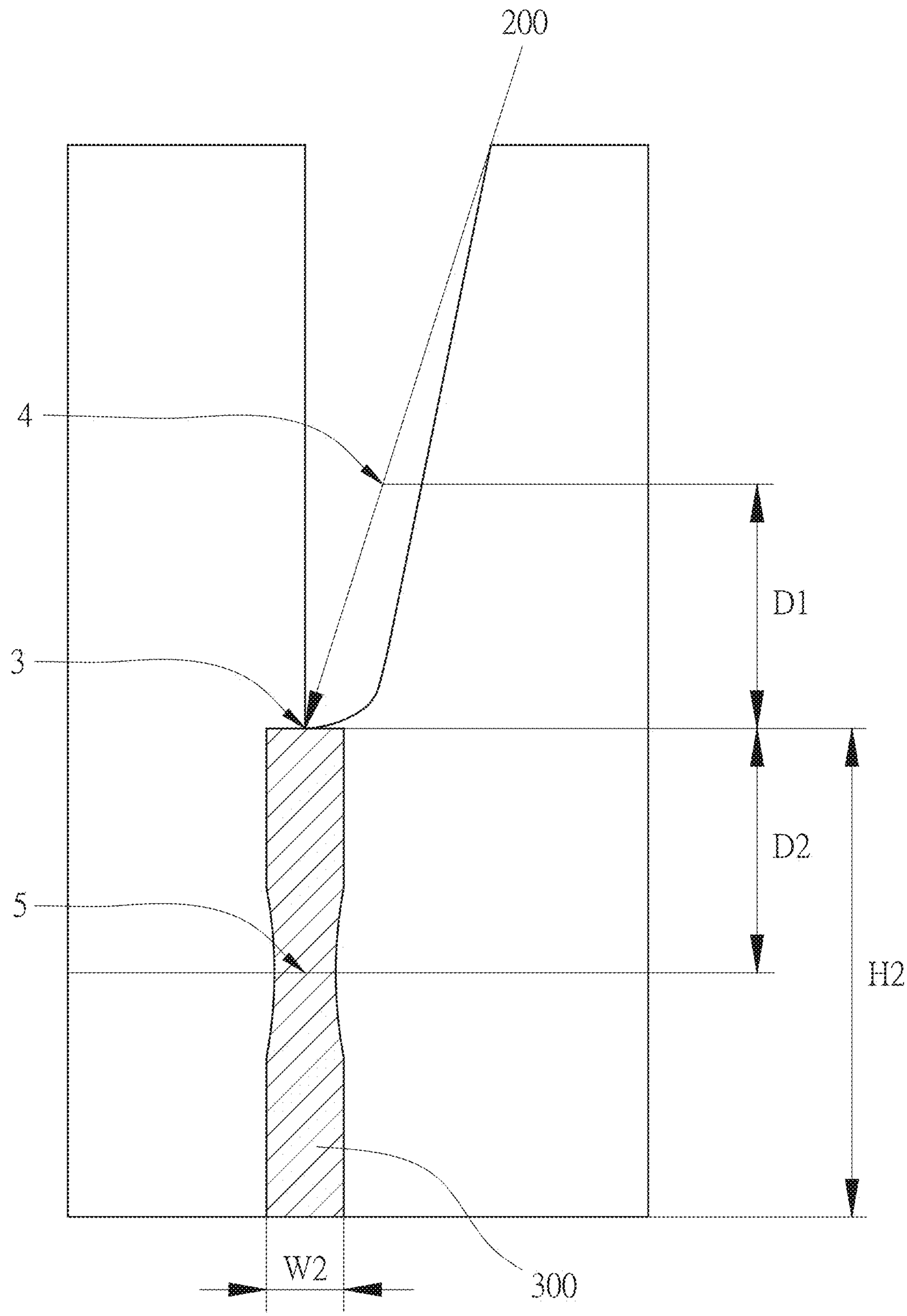

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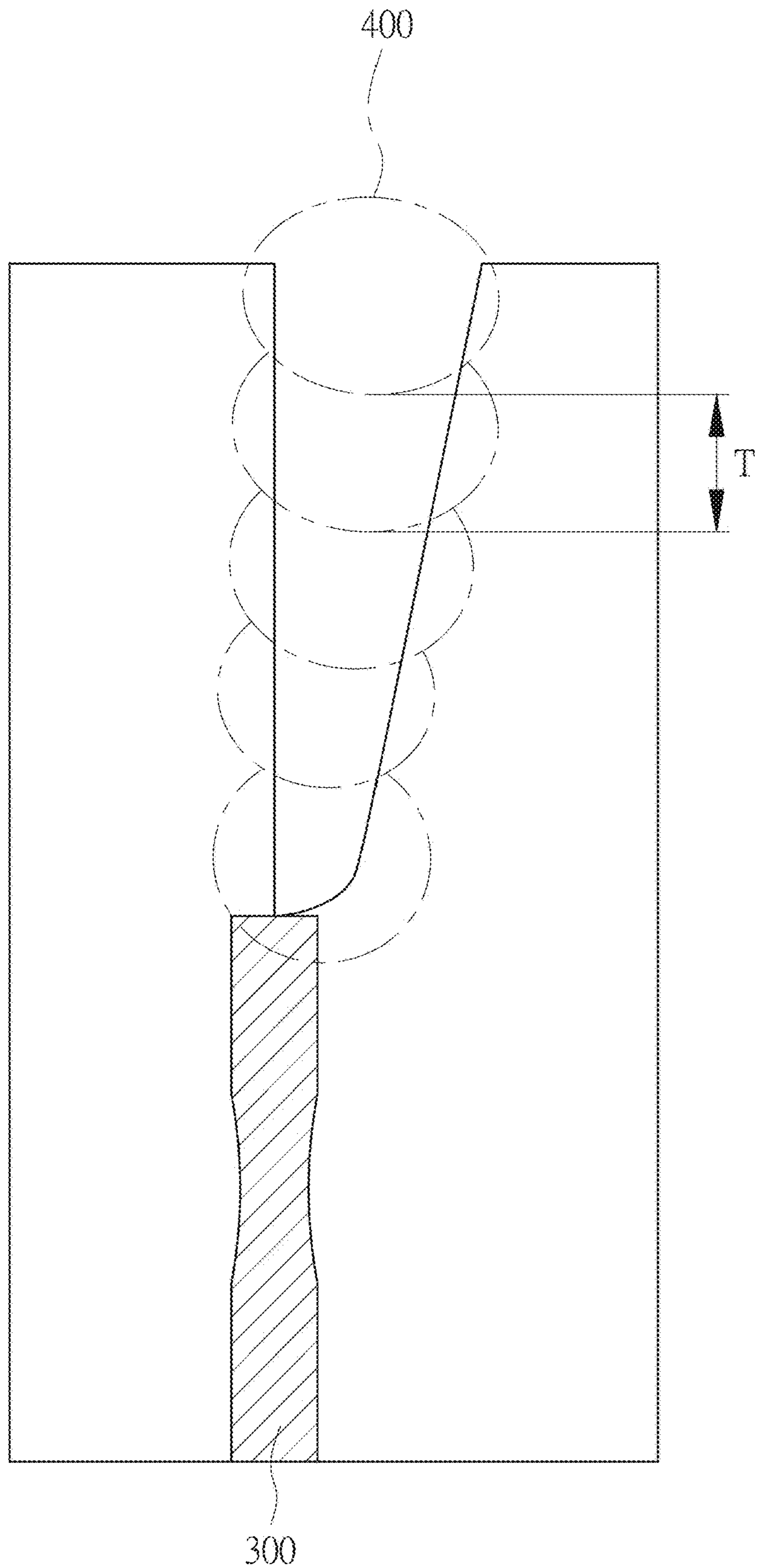

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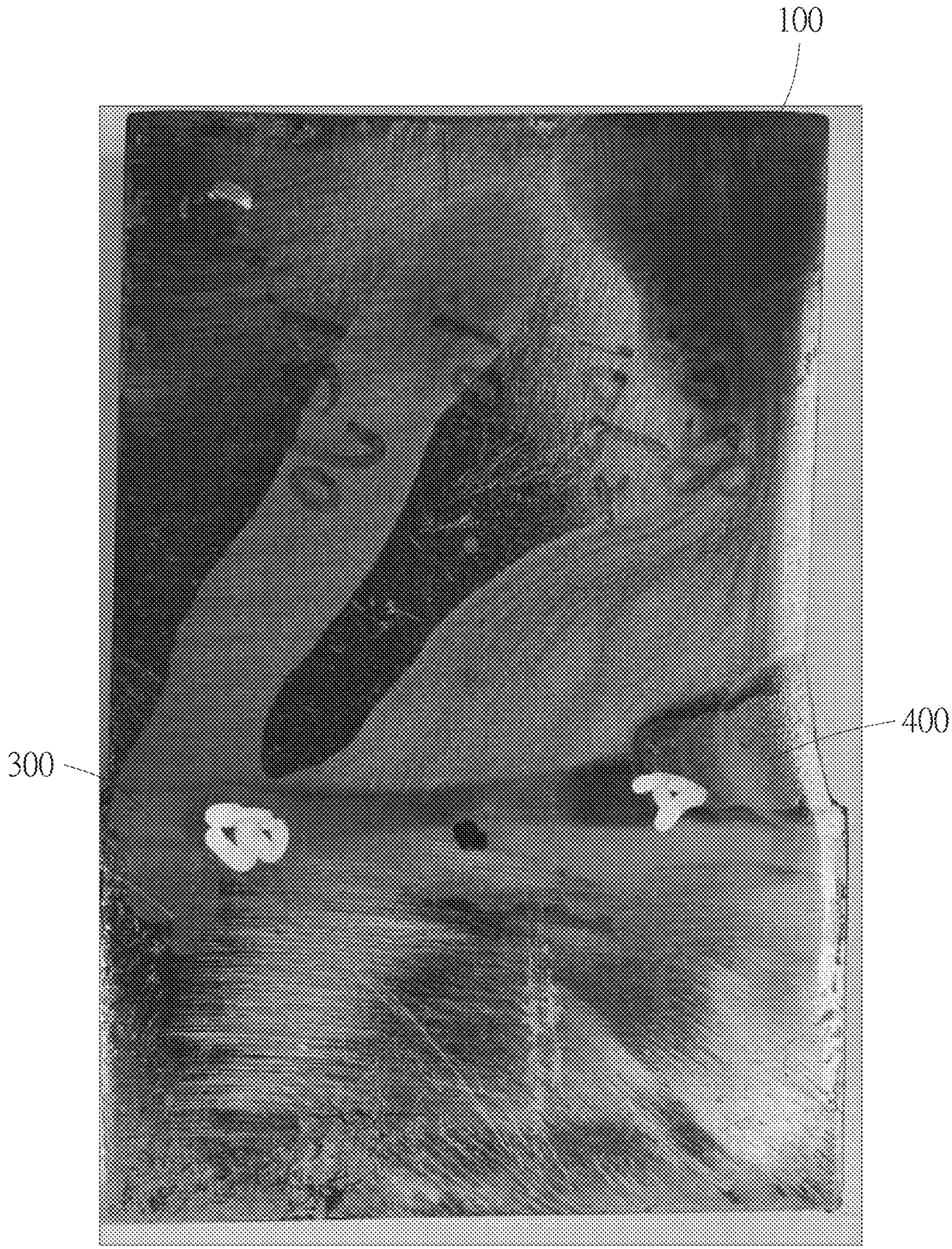

FIG. 7

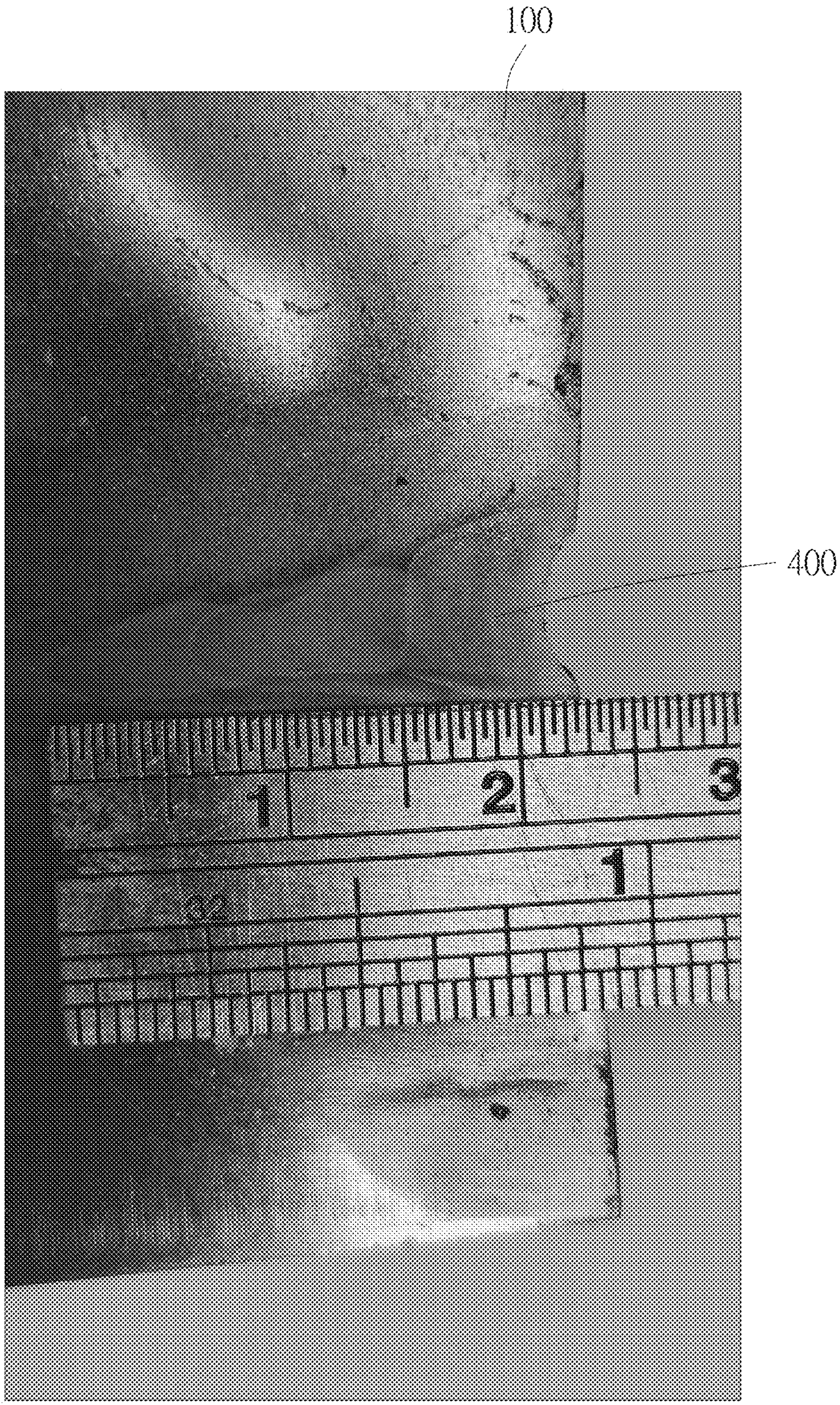

FIG. 8

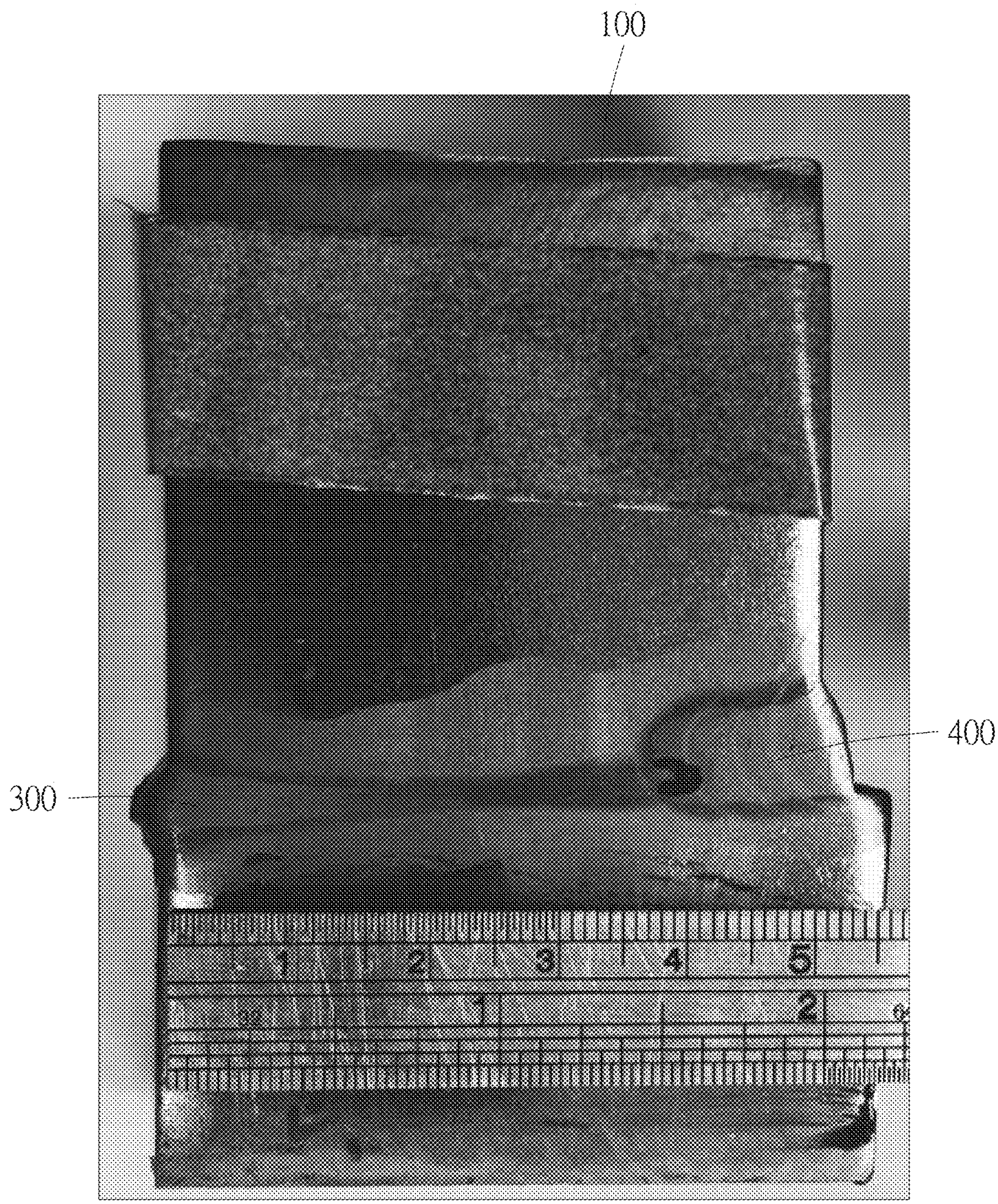

FIG. 9

THICK STEEL PLATE ASSEMBLY WELDED BY LASER WELDING AND LASER WELDING METHOD THEREOF

FIELD OF THE INVENTION

[0001] The present invention relates to a thick steel plate assembly welded by laser welding and a laser welding method for thick steel plates.

BACKGROUND OF THE INVENTION

[0002] Using a laser to weld thick steel plates is a common technology nowadays. For example, Chinese Patent Publication No. CN 113695744 A discloses an energy-constrained, narrow-gap, wire-filling laser welding method, relating to the field of laser welding technology. The method includes the following steps: having a groove design, wherein a groove is formed between two parts to be welded; preparation before welding, wherein the two parts to be welded are fixed on the workbench, and a protective gas is used to clean the groove; pre-welding, wherein a welding wire is fed into the bottom of the groove, and a laser beam pre-welds the welding wire and the parts to be welded for the welding wire to be fixed at the bottom of the groove, and pre-welding uses a positive defocused laser beam to enter the groove for welding; continuous welding, wherein a negative defocused laser beam enters the groove for welding, and the focal point is located inside the welding wire.

[0003] However, in the aforementioned patent, it is necessary to machine bevels on both parts to be welded to form the groove. The process is more troublesome and the cost is higher.

SUMMARY OF THE INVENTION

[0004] According to one aspect of the present invention, a laser welding method for thick steel plates is provided. The laser welding method comprises the following steps of: joining two thick steel plates, wherein a joint face of the thick steel plates extends along a joint line, at least one portion of the joint face of one of the thick plate steels is formed with a groove along a grooving line, a grooving angle is defined between the grooving line and the joint line, the grooving angle is between 3 degrees and 15 degrees, and a bottom width of the groove is between R0 and R10 mm; feeding a filler into the groove and using a laser to weld the joint face having the groove through the filler to form a filler region, wherein the laser has a power of between 7,000 and 20,000 watts. After the filler region is cooled and solidified, the thick steel plates are combined with each other. A thickness of each filler layer in the filler region is between 4 and 6 mm.

[0005] According to another aspect of the present invention, a thick steel plate assembly welded by laser welding is provided. The thick steel plate assembly comprises two thick steel plates and a filler region. The steel plates are joined. A joint face of the steel plates extends along a joint line. At least one portion of the joint face of one of the thick plate steels is formed with a groove along a grooving line. A grooving angle is defined between the grooving line and the joint line. The grooving angle is between 3 degrees and 15 degrees. A bottom width of the groove is between R0 and R10 mm. The filler region is formed by feeding a filler into the groove and using a laser to weld the joint face having the groove through the filler. The laser has a power of between 7,000 and 20,000 watts. After the filler region is cooled and solidified, the thick steel plates are combined with each other. A thickness of each filler layer in the filler region is between 4 and 6 mm.

[0006] Preferably, one portion not having the groove of the joint face of one of the thick plate steels has a depth greater than 0 mm but not greater than 50 mm. Before using the laser to weld the joint face having the groove through the filler, the joint face not having the groove is welded using the laser. A welding angle is defined between the laser and the joint line. The welding angle is between 0 degrees and 10 degrees. The laser welds the joint face to form a molten region extending along the joint line. A molten depth of the molten region is substantially equal to the depth of the joint face not having the groove. A molten width of the molten region is between 4 and 7 mm. After the molten region and the filler region are cooled and solidified, the thick steel plates are combined with each other.

[0007] Preferably, when the laser is used to weld the joint face not having the groove, at least one of the following conditions is met: a defocusing distance of the laser is between 0 and ±H/2 mm, H is the depth of the joint face not having the groove; the laser has a power variation of ±3%, a wavelength of the laser is between 1030 and 1080 nanometers; the laser has a welding speed ranging from 5 to 80 millimeters per second; when the thick steel plates are joined, the thick steel plates have a clearance range of not more than 5% of a width of the thick steel plates; the power of the laser is between 20,000 and 30,000 watts; a protective gas is provided to remove a plasma generated during welding of the laser; the protective gas is one of argon, helium, nitrogen and carbon dioxide, or a combination thereof.

[0008] Preferably, an action of the laser is one of scanning, robotic arc scanning and swing scanning. The laser has a swing frequency ranging from 0 to 100 Hz.

[0009] Preferably, when the laser is used to weld the joint face having the groove through the filler, at least one of the following conditions is met: the laser has a welding speed ranging from 5 to 30 millimeters per second; the filler is fed at a feeding speed ranging from 10 to 200 millimeters per second; a feeding angle is defined between the filler and the joint line and the feeding angle is between 0 degrees and 60 degrees; the bottom width is a diameter of a circle at a bottom of the groove.

[0010] According to the above technical features, the present invention can achieve the following effects:

[0011] Since only one of the thick plate steels is formed with the groove, the process can be simplified greatly and the cost can be reduced.

[0012] The bottom of the groove has a circle to ensure that there is no dead space for subsequent filling.

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] FIG. **1** is an exploded view of an embodiment the present invention;

[0014] FIG. **2** is a flow block diagram of the embodiment of the present invention;

[0015] FIG. **3** is a front view of the embodiment of the present invention, showing the groove;

[0016] FIG. **4** is a first schematic view of the embodiment of the present invention, showing the welding angle;

[0017] FIG. **5** is a second schematic view of the embodiment of the present invention, showing the defocusing distance;

[0018] FIG. **6** is a third schematic view of the embodiment of the present invention, showing the filler region;

[0019] FIG. **7** is a first photograph of the embodiment of the present invention;

[0020] FIG. **8** is a second photograph of the embodiment of the present invention; and

[0021] FIG. **9** is a third photograph of the embodiment of the present invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0022] Embodiments of the present invention will now be described, by way of example only, with reference to the accompanying drawings.

[0023] As shown in FIG. **1** through FIG. **3**, the present invention discloses a thick steel plate assembly welded by laser welding and a laser welding method for thick steel plates.

[0024] The laser welding method for thick steel plates comprises the following steps:

[0025] Two thick steel plates **100** are joined. A joint face **1** of the thick steel plates **100** extends along a joint line L. When the thick steel plates **100** are joined, the thick steel plates **100** have a clearance range of not more than 5% of the width of the thick steel plates **100**.

[0026] At least one portion of the joint face **1** of one of the thick plate steels **100** is formed with a groove **2** along a grooving line L**2**. A grooving angle θ**1** is defined between the grooving line L**2** and the joint line L**1**. The grooving angle θ**1** is between 3 degrees and 15 degrees. The bottom width W**1** of the groove **2** is between R0 and R10 mm. The depth (i.e., the first depth H**1**) of the groove **2** is between 10 and 50 mm. In actual implementation, the bottom width W**1** may be the diameter of a circle at the bottom of the groove **2** to ensure that there is no dead space for subsequent filling.

[0027] Please refer to FIG. **2** through FIG. **5**. One portion not having the groove **2**, (i.e., the second depth H**2**) of the joint face **1** of one of the thick plate steels **100** has a depth greater than 0 mm but not greater than 50 mm. The joint face **1** not having the groove **2** is first welded using a laser **200**. A welding angle θ**2** is defined between the laser **200** and the joint line L. The welding angle θ**2** is between 0 degrees and 10 degrees.

[0028] The laser **200** welds the joint face **1** to form a molten region **300** extending along the joint line L. The molten depth of the molten region **300** is substantially equal to the depth of the joint face **1** not having the groove **2**. The molten width W of the molten region **300** is between 4 and 7 mm.

[0029] A defocusing distance of the laser **200** is between 0 and ±25 mm. That is, the positive defocusing distance D**1** is up to 25 mm, and the negative defocusing distance D**2** is up to −25 mm, which is equivalent to half of the second depth H**2**, namely, the depth of the joint face not having the groove **2**. If the surface of the thick steel plates **100** serves as the focal point **3**, the positive defocusing point **4** will be 25 mm above the focal point **3**, and the negative defocusing point **5** will be 25 mm below the focal point **3**.

[0030] The laser **200** has a power variation of ±3%. The wavelength of the laser **200** is between 1030 and 1080 nanometers. The laser **200** has a welding speed ranging from 5 to 80 millimeters per second. The power of the laser **200** is between 20,000 and 30,000 watts.

[0031] The above-mentioned values may be changed according to the second depth H**2**. For example, when the second depth H**2** is between 40 and 50 mm, the power of the laser **200** may be 30,000 watts, and the welding speed is between 5 and 80 millimeters per second.

[0032] A protective gas is used to remove a plasma generated during the welding of the laser **200**. The protective gas is one of argon, helium, nitrogen and carbon dioxide, or a combination thereof.

[0033] By changing the defocusing distance and the power variation, the laser **200** enables the joint face **1** to form the molten region **300** extending along the joint line L.

[0034] Please refer to FIG. **6** and FIG. **7**, in cooperation with FIG. **2** through FIG. **4**. Next, a filler is fed into the groove **2**, and the laser **200** welds the joint face **1** having the groove **2** through the filler to form a filler region. The thickness T of each filler layer in the filler region **400** is between 4 and 6 mm.

[0035] The power of the laser **200** is between 7,000 and 20,000 watts. The welding speed of the laser **200** is between 5 and 30 millimeters per second. The filler is fed at a feeding speed ranging from 10 to 200 millimeters per second. A feeding angle is defined between the filler and the joint line L**1**. The feeding angle is between 0 degrees and 60 degrees.

[0036] The action of the laser **200** is one of scanning, robotic arc scanning and swing scanning. The laser **200** has a swing frequency ranging from 0 to 100 Hz.

[0037] After the molten region **300** and the filler region **400** are cooled and solidified, the thick steel plates **100** are combined with each other to form a thick steel plate assembly.

[0038] Please refer to FIG. **8** and FIG. **9**, in cooperation with FIG. **3** and FIG. **6**. As can be seen from the photographs of the actual thick steel plate assembly manufactured by the laser welding method, the thickness T of each filler layer in the filler region **400** is indeed between 4 and 6 mm, the first depth H**1** is indeed between 10 and 50 mm, and the second depth H**2** is indeed not greater than 50 mm.

[0039] Since only one of the thick plate steels **100** is formed with the groove **2**, the process can be simplified greatly and the cost can be reduced.

[0040] Although particular embodiments of the present invention have been described in detail for purposes of illustration, various modifications and enhancements may be made without departing from the spirit and scope of the present invention. Accordingly, the present invention is not to be limited except as by the appended claims.

What is claimed is:

1. A laser welding method for thick steel plates, comprising the following steps of:

joining two thick steel plates, wherein a joint face of the thick steel plates extends along a joint line, at least one portion of the joint face of one of the thick plate steels is formed with a groove along a grooving line, a grooving angle is defined between the grooving line and the joint line, the grooving angle is between 3 degrees and 15 degrees, and a bottom width of the groove is between R0 and R10 mm;

feeding a filler into the groove and using a laser to weld the joint face having the groove through the filler to form a filler region, wherein the laser has a power of between 7,000 and 20,000 watts;

wherein after the filler region is cooled and solidified, the thick steel plates are combined with each other, and a thickness of each filler layer in the filler region is between 4 and 6 mm.

2. The laser welding method as claimed in claim **1**, wherein one portion not having the groove of the joint face of one of the thick plate steels has a depth greater than 0 mm but not greater than 50 mm, before using the laser to weld the joint face having the groove through the filler, the joint face not having the groove, is first welded using the laser, a welding angle is defined between the laser and the joint line, the welding angle is between 0 degrees and 10 degrees; the laser welds the joint face to form a molten region extending along the joint line, a molten depth of the molten region is substantially equal to the depth of the joint face not having the groove, a molten width of the molten region is between 4 and 7 mm, after the molten region and the filler region are cooled and solidified, the thick steel plates are combined with each other.

3. The laser welding method as claimed in claim **2**, wherein when the laser is used to weld the joint face not having the groove, at least one of the following conditions is met: a defocusing distance of the laser is between 0 and ±H/2 mm, H is the depth of the joint face not having the groove; the laser has a power variation of ±3%, a wavelength of the laser is between 1030 and 1080 nanometers; the laser has a welding speed ranging from 5 to 80 millimeters per second; when the thick steel plates are joined, the thick steel plates have a clearance range of not more than 5% of a width of the thick steel plates; the power of the laser is between 20,000 and 30,000 watts; a protective gas is provided to remove a plasma generated during welding of the laser; the protective gas is one of argon, helium, nitrogen and carbon dioxide, or a combination thereof.

4. The laser welding method as claimed in claim **1**, wherein an action of the laser is one of scanning, robotic arc scanning and swing scanning, and the laser has a swing frequency ranging from 0 to 100 Hz.

5. The laser welding method as claimed in claim **1**, wherein when the laser is used to weld the joint face having the groove through the filler, at least one of the following conditions is met: the laser has a welding speed ranging from 5 to 30 millimeters per second; the filler is fed at a feeding speed ranging from 10 to 200 millimeters per second; a feeding angle is defined between the filler and the joint line and the feeding angle is between 0 degrees and 60 degrees; the bottom width is a diameter of a circle at a bottom of the groove.

6. A thick steel plate assembly welded by laser welding, comprising:

two thick steel plates, the steel plates being joined, a joint face of the steel plates extending along a joint line, at least one portion of the joint face of one of the thick plate steels being formed with a groove along a grooving line, a grooving angle being defined between the grooving line and the joint line, the grooving angle being between 3 degrees and 15 degrees, a bottom width of the groove being between R0 and R10 mm;

a filler region, the filler region being formed by feeding a filler into the groove and using a laser to weld the joint face having the groove through the filler, the laser having a power of between 7,000 and 20,000 watts;

wherein after the filler region is cooled and solidified, the thick steel plates are combined with each other, and a thickness of each filler layer in the filler region is between 4 and 6 mm.

7. The thick steel plate assembly as claimed in claim **6**, further comprising a molten region adjacent to the filler region, wherein the molten region extends along the joint line, one portion not having the groove of the joint face of one of the thick plate steels has a depth greater than 0 mm but not greater than 50 mm, a molten depth of the molten region is substantially equal to the depth of the joint face not having the groove, a molten width of the molten region is between 4 and 7 mm, before using the laser to weld the joint face having the groove through the filler, the molten region is formed by welding the joint face not having the groove using the laser, a welding angle is defined between the laser and the joint line, the welding angle is between 0 degrees and 10 degrees, after the molten region and the filler region are cooled and solidified, the thick steel plates are combined with each other.

8. The thick steel plate assembly as claimed in claim **7**, wherein when the laser is used to weld the joint face not having the groove, at least one of the following conditions is met: a defocusing distance of the laser is between 0 and ±H/2 mm, H is the depth of the joint face not having the groove; the laser has a power variation of ±3%, a wavelength of the laser is between 1030 and 1080 nanometers; the laser has a welding speed ranging from 5 to 80 millimeters per second; when the thick steel plates are joined, the thick steel plates have a clearance range of not more than 5% of a width of the thick steel plates; the power of the laser is between 20,000 and 30,000 watts; a protective gas is provided to remove a plasma generated during welding of the laser; the protective gas is one of argon, helium, nitrogen and carbon dioxide, or a combination thereof.

9. The thick steel plate assembly as claimed in claim **6**, wherein an action of the laser is one of scanning, robotic arc scanning and swing scanning, and the laser has a swing frequency ranging from 0 to 100 Hz.

10. The thick steel plate assembly as claimed in claim **6**, wherein when the laser is used to weld the joint face having the groove through the filler, at least one of the following conditions is met: the laser has a welding speed ranging from 5 to 30 millimeters per second; the filler is fed at a feeding speed ranging from 10 to 200 millimeters per second; a feeding angle is defined between the filler and the joint line and the feeding angle is between 0 degrees and 60 degrees; the bottom width is a diameter of a circle at a bottom of the groove.

* * * * *